

Fire resistance test report

Issuing laboratory: Warringtonfire Testing and Certification Limited

Test standard: BS EN 1365-1:2012

Test sponsor: earth4Earth

Sheffield Technology Parks,
Sheffield,
S1 2NS

Product: Symmetrical, Loadbearing Wall Assembly

Report number: 557960/R

Test date: 05 May 2026

Issue: 1

Warringtonfire, accredited for compliance with ISO/IEC 17025:2017 – Testing

Registered Laboratory:

Warringtonfire
Holmesfield Road
Warrington
Cheshire
WA1 2DS
United Kingdom



Approved Body Number 0833



Summary of Tested Specimen

The specimen had overall nominal dimensions of 2785 mm high by 3015 mm wide by 215 mm thick and was formed from two courses of earth4Earth L10 Low Carbon Brick with dimensions of 65 mm deep x 102 mm wide x 215 mm long.

The brick was lay in an English bond pattern and Bonded together (hand trowelled) with mortar using Blue Circle Mastercrete cement and Beasley & Fildes Ltd sand.

At the request of the test sponsor, the wall supported a total applied load of 30 kN. The load was applied via steel load spreader and hydraulic rams to the underside of the test specimen. The wall was unrestrained on both vertical edges.

Detailed drawings of the test specimen(s) and a comprehensive description of the test construction based on a detailed survey of the specimen(s) and information supplied by the sponsor of the test are included in the Test Specimen and Schedule of Components sections of this report.

Test Results

Loadbearing Capacity	The maximum allowable vertical contraction and the maximum rate of vertical contraction for the specimen, as specified by the Standard, are calculated as 27.85 mm and 8.36 mm per minute respectively. The deflection is taken at two locations with the average used to calculate the loadbearing capacity. The requirements were satisfied for:																			
	264 minutes	No failure*																		
Integrity	It is required that the specimen retains its separating function, without: - causing ignition of a cotton pad when applied - permitting the penetration of a gap gauge as specified in BS EN 1363-1: 2020 - sustained flaming on the unexposed surface - subsequent failure of loadbearing capacity These requirements were satisfied for the periods shown below: <table><tr><td>Sustained flaming</td><td>264 minutes</td><td colspan="3">No failure*</td></tr><tr><td>Gap gauge</td><td>264 minutes</td><td colspan="3">No failure*</td></tr><tr><td>Cotton pad</td><td>264 minutes</td><td colspan="3">No failure*</td></tr></table>					Sustained flaming	264 minutes	No failure*			Gap gauge	264 minutes	No failure*			Cotton pad	264 minutes	No failure*		
Sustained flaming	264 minutes	No failure*																		
Gap gauge	264 minutes	No failure*																		
Cotton pad	264 minutes	No failure*																		
Insulation	It is required that the mean temperature rise of the unexposed surface shall not be greater than 140°C and that the maximum temperature rise shall not be greater than 180°C. Insulation failure also occurs simultaneously with integrity failure. These requirements were satisfied for the period shown below:																			
	264 minutes	No failure*																		
Radiation	BS EN 1363-2: 1999 requires that the time for the measured radiation to exceed 5, 10, 15, 20 and 25 kW/m ² is reported.																			
	5 kW/m ²	10 kW/m ²	15 kW/m ²	20 kW/m ²	25 kW/m ²															
	264 minutes*	264 minutes*	264 minutes*	264 minutes*	264 minutes*															
	*Test was discontinued after a period of 264 minutes.																			

Date of Test 05 May 2026

Location of Test Element Materials Technology, 722 Birchwood Park, Warrington WA3 6FW, United Kingdom

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Quality Management

Issue No: 1		Issue Date: 24 June 2026	
Responsible Officer:	L. Doyle* Technical Officer	Approved By:	K. Brennan* Report Co-ordinator / Technical Officer
			

* For and on behalf of **Warringtonfire**.

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Test Conditions

Standard	In accordance with BS EN 1365-1:2012 Fire resistance tests for loadbearing elements - Part 1: Walls.
Sampling	Warringtonfire was not involved in the sampling or selection of the tested specimen or any of the components. The results obtained during the test only apply to the test samples as received and tested by Warringtonfire.
Installation	The specimen was received on the 24 March 2026 and installed into a refractory concrete lined steel restraint frame, such that both vertical edges had freedom of movement. The specimen was installed by representatives of WarringtonFire between the 30 March 2026 and 02 April 2026.
Specimen Direction	Tested specimen was symmetrical . The Test Sponsor defined the direction in which the specimen was tested, and it is shown in the test specimen drawings section of this report.
Conditioning	The specimen's storage, construction, and test preparation took place in the test laboratory over a total, combined time of 43 days. Throughout this period of time both the temperature and the humidity of the laboratory were measured and recorded as being within a range of from 7.0°C to 26.5°C and 26.5% to 80.0% respectively.
Instruction to Test	The test was conducted on the 05 May 2026 at the request of earth4Earth, the Test Sponsor .
Ambient Temperature	The ambient air temperature in the vicinity of the test construction was 17°C at the start of the test with a maximum variation of +3°C during the test.
Furnace	The furnace was controlled so that its mean temperature complied with the requirements of BS EN 1363-1: 2020 Clause 5.1 using nine plate thermometers, distributed over a plane 100 mm from the surface of the test construction.
Thermocouples	Thermocouples were provided to monitor the unexposed surface of the specimen. The output of all instrumentation was recorded at no less than one minute intervals. The locations and reference numbers of the various unexposed surface thermocouples are shown in Figure 1.
Application of the load	The full test load was applied via hydraulic rams uniformly distributed over the test Specimen 30 minutes before the commencement of the test.
Radiation	A water-cooled foil heat-flux meter was used to record the heat radiation from the specimen. The heat flux meter was positioned at a distance of 1 metre from the centre of the specimen.
Furnace Pressure	After the first five minutes of testing and for the remainder of the test, the furnace atmospheric pressure was controlled so that it complied with the requirements of BS EN 1363-1: 2020, clause 5.2.1 The calculated pressure differential relative to the laboratory atmosphere at the top of the specimen was 19.2 ($\pm$ 5) Pa between 5 and 10 minutes and 19.2 ($\pm$ 3) Pa thereafter.

Test Specimen Drawings

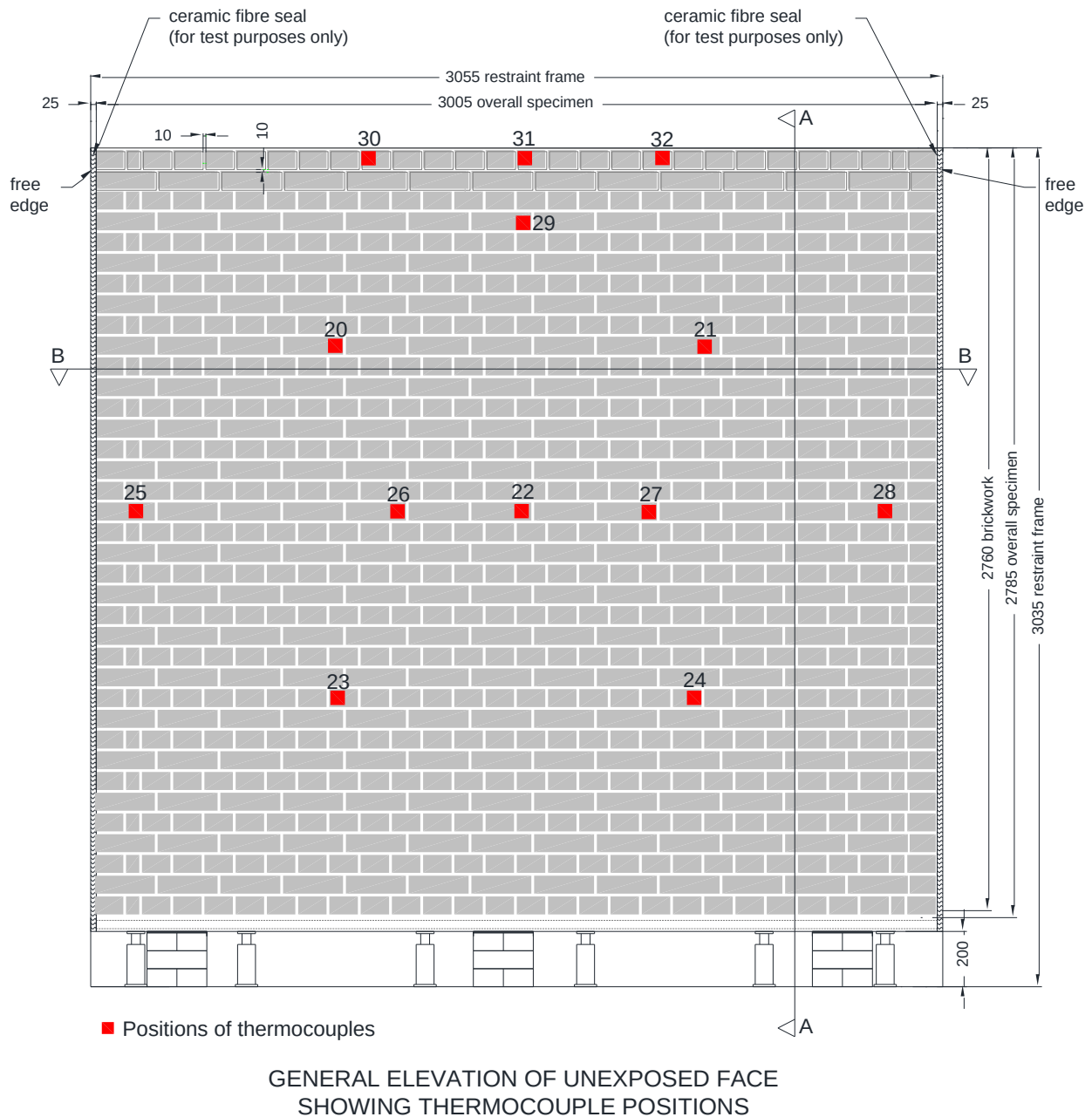
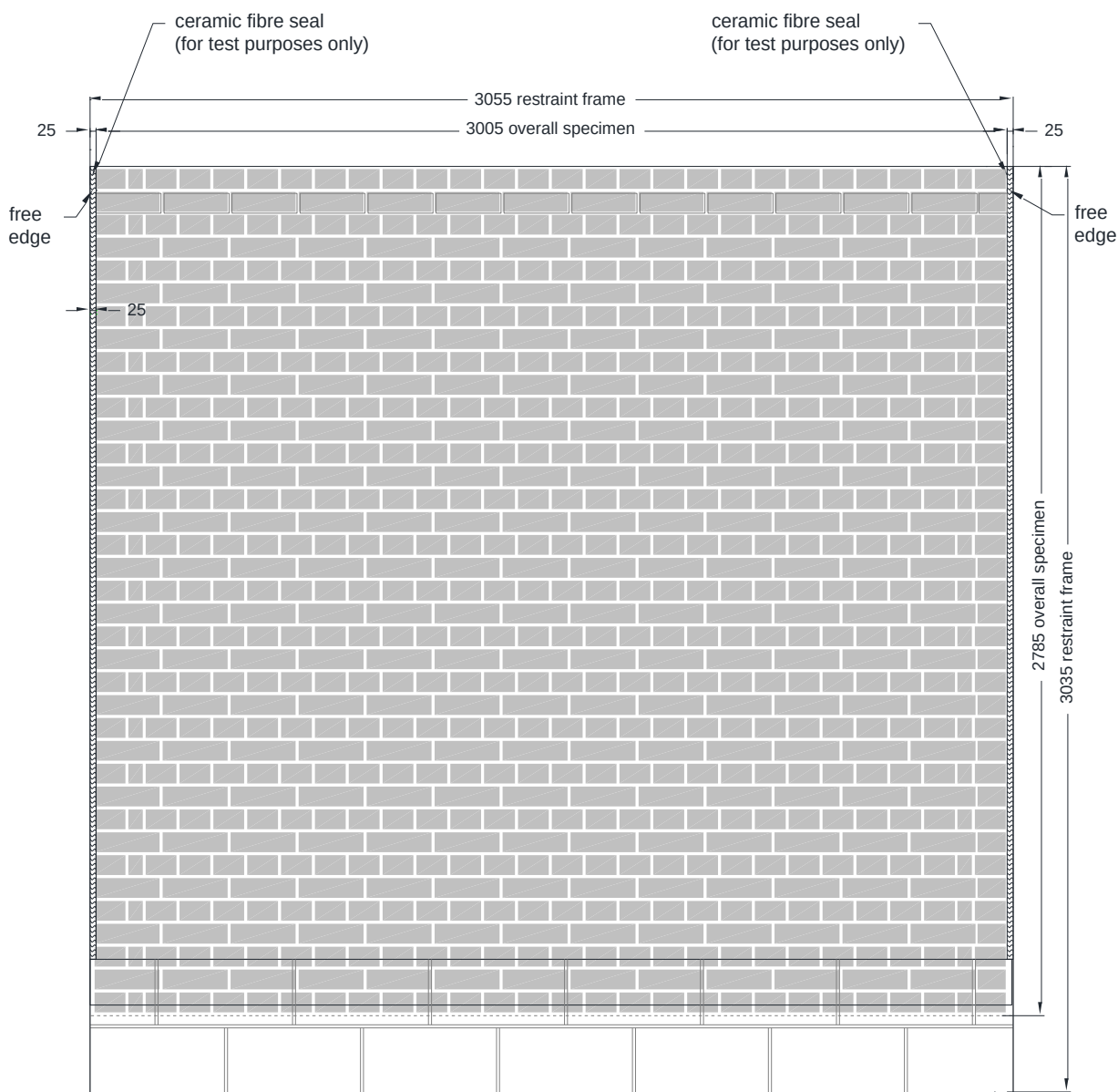


Figure 1. General Elevation of Unexposed Face Showing Thermocouple Positions



GENERAL ELEVATION OF EXPOSED FACE

Figure 2. General Elevation of Exposed Face

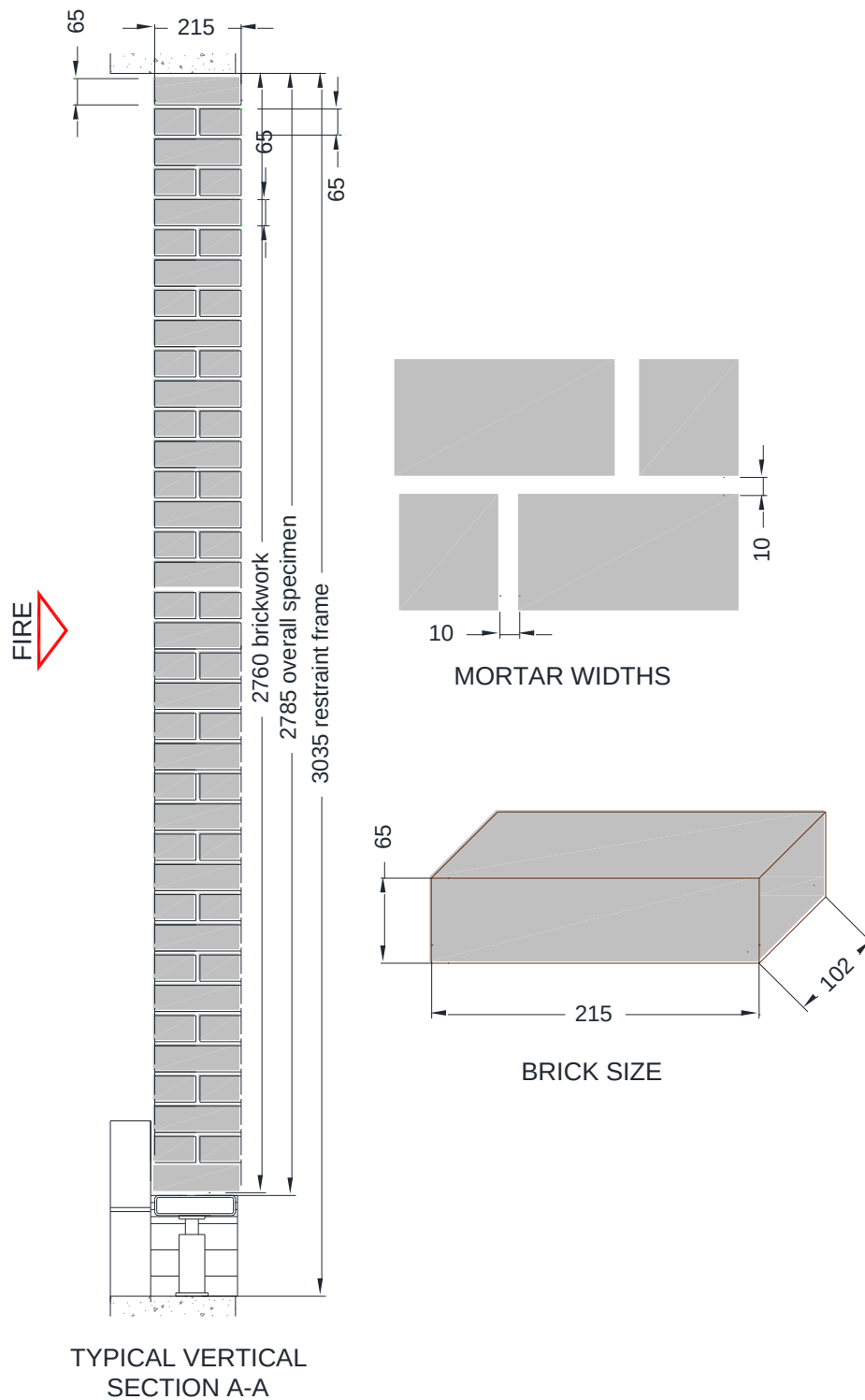


Figure 3. Typical Vertical Section A-A

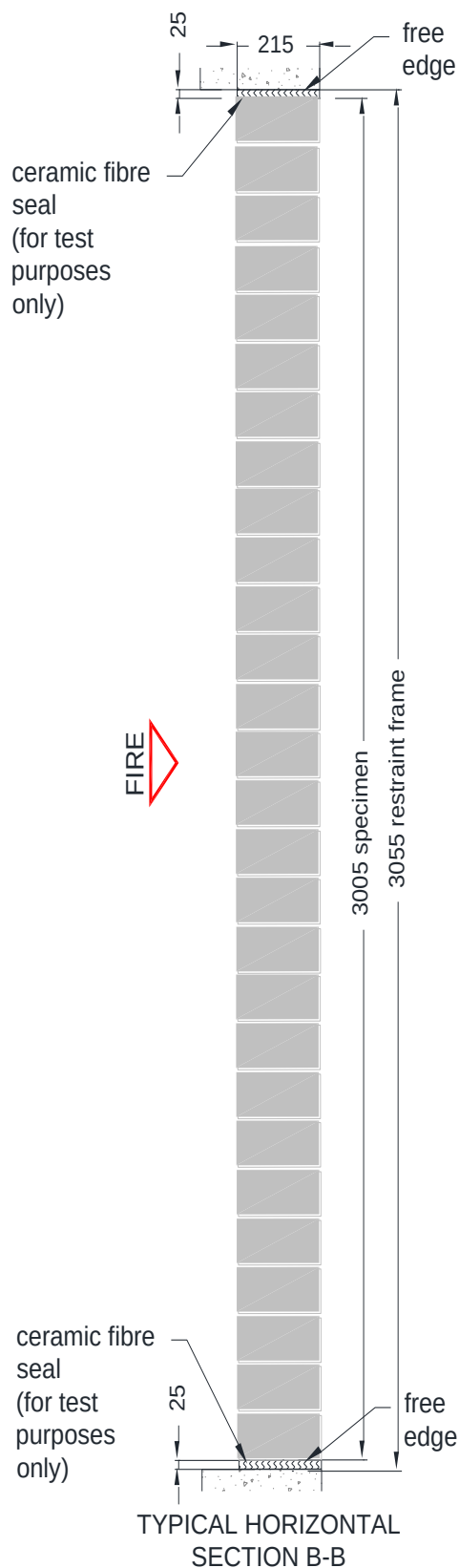


Figure 4. Typical Horizontal Section B-B

Schedule of Components

Table 1. details the schedule of components which describes the test specimen and lists the components used in the construction of the test specimen. These were provided by the test sponsor and surveyed by Warringtonfire.

All measurements were verified by Warringtonfire unless stated otherwise in the schedule of components. All components marked with an “*” have not been verified by Warringtonfire.

Table 1. Schedule of components

Test Construction

Concrete Bricks	
Manufacturer	earth4Earth
Reference	e4E-2025/10-L10132-02 ; e4E-2025/10-L10132-03 ; e4E-2025/10-L10132-04
Material	L10 Low Carbon Brick
Density	1900 kg/m ³ (stated)
Overall size	65 mm deep x 102 mm wide x 215 mm long
Laying method	Bonded together (hand trowelled) with mortar using Blue Circle Mastercrete cement and Beasley & Fildes LTD sand

Fire Stopping (Supplied by WarringtonFire)

High temperature insulation wool	
Manufacturer	Morgan Advanced Materials
Reference	Superwool Plus
Material	Alkaline Earth Silicate Fibre Based Insulation
Thickness	20 mm, uncompressed
Density	96 kg/m ³ (stated)

Test Observations

Time All observations are from the unexposed face unless noted otherwise.

mins secs

-30	00	Load applied and maintained.
00	00	The Test Commences.
19	10	Moisture spots appearing through mortar on the unexposed face.
24	00	No significant visible change to the exposed face.
26	00	More moisture spots developing in the mortar on the unexposed face.
35	00	Steam/smoke release from the head of the unexposed face.
40	00	No significant visible change to the exposed face.
49	10	Moisture release continuing on the unexposed face.
74	00	No significant visible change to the exposed face.
88	00	Steam/smoke release through areas in the mortar on the unexposed face.
125	00	No significant visible change to the unexposed face.
132	00	No significant visible change to the exposed face.
177	00	No significant visible change to the exposed face.
178	00	Moisture now drying on the unexposed face.
232	00	No significant visible change to the unexposed face.
253	00	No significant visible change to the exposed face.
264	00	Test discontinued at sponsor's request.

Test Photographs

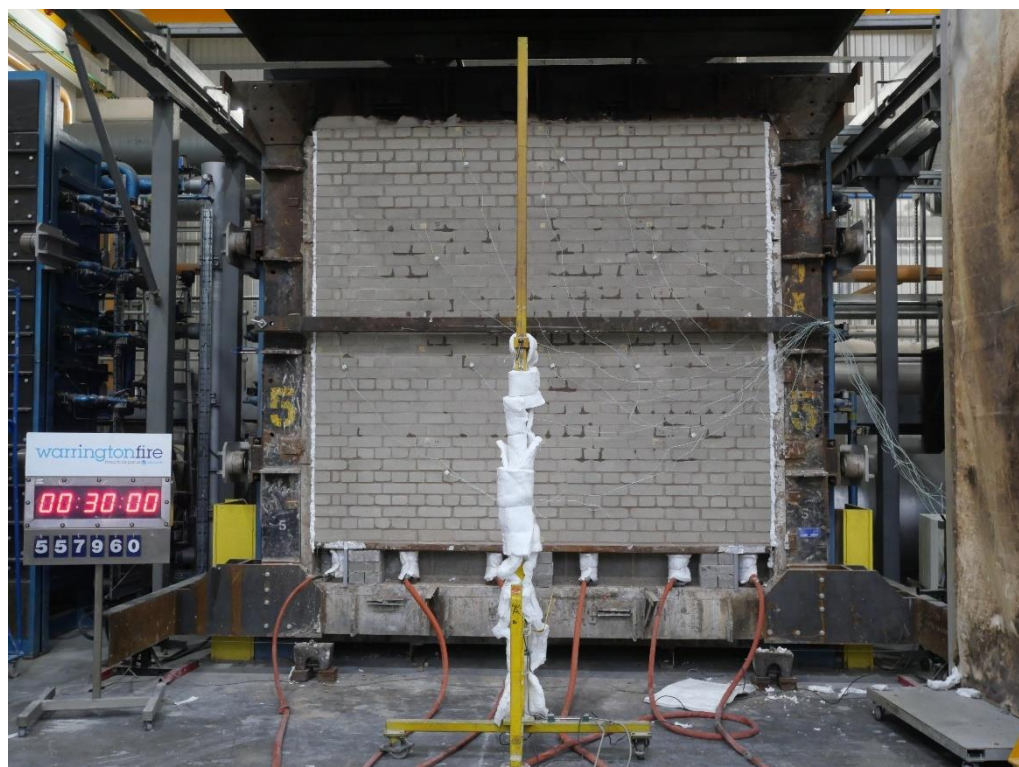
The exposed face of the wall assembly prior to testing



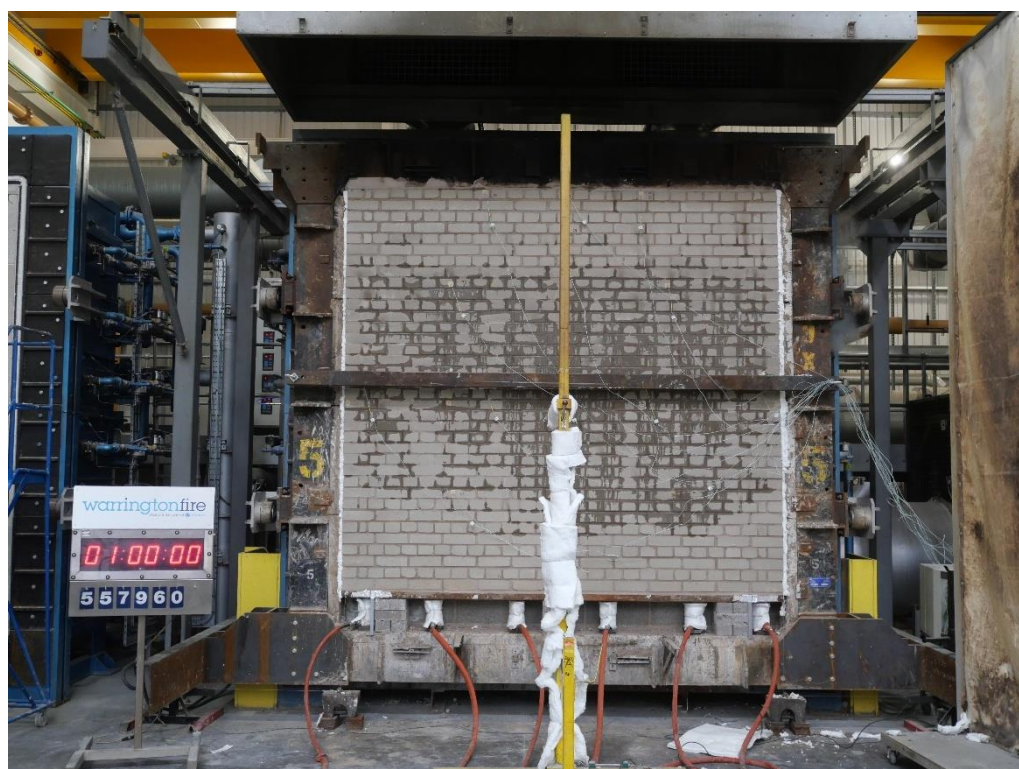
The unexposed face of the wall assembly prior to the start of the test



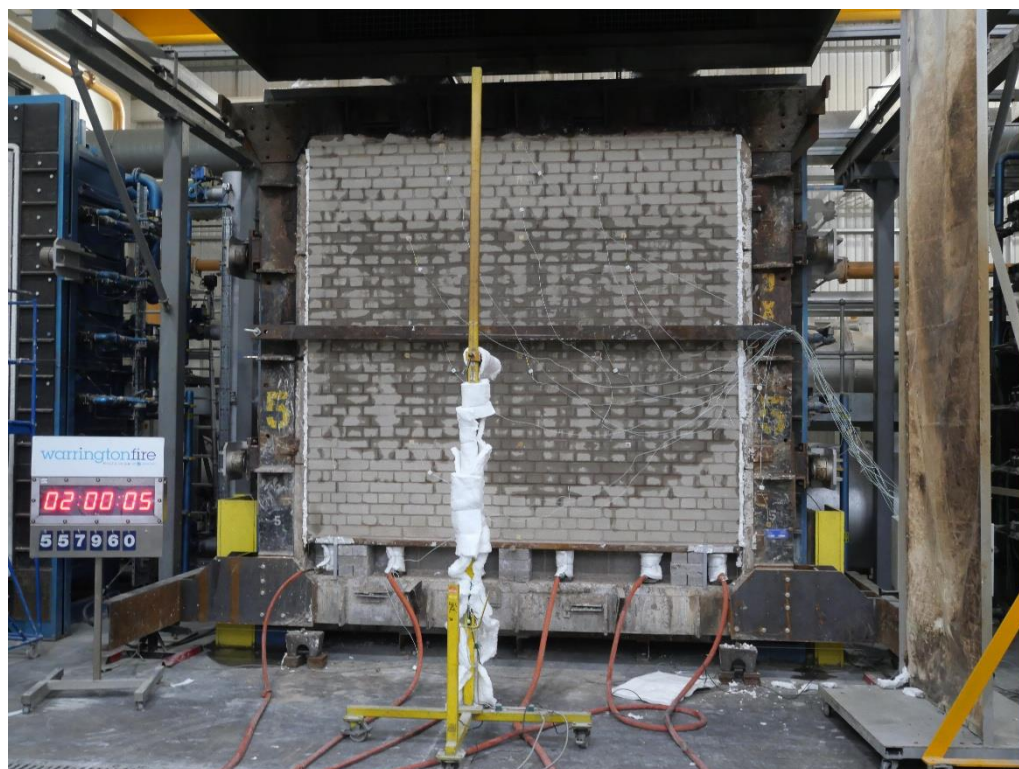
The unexposed face of the wall assembly after a test duration of 30 minutes



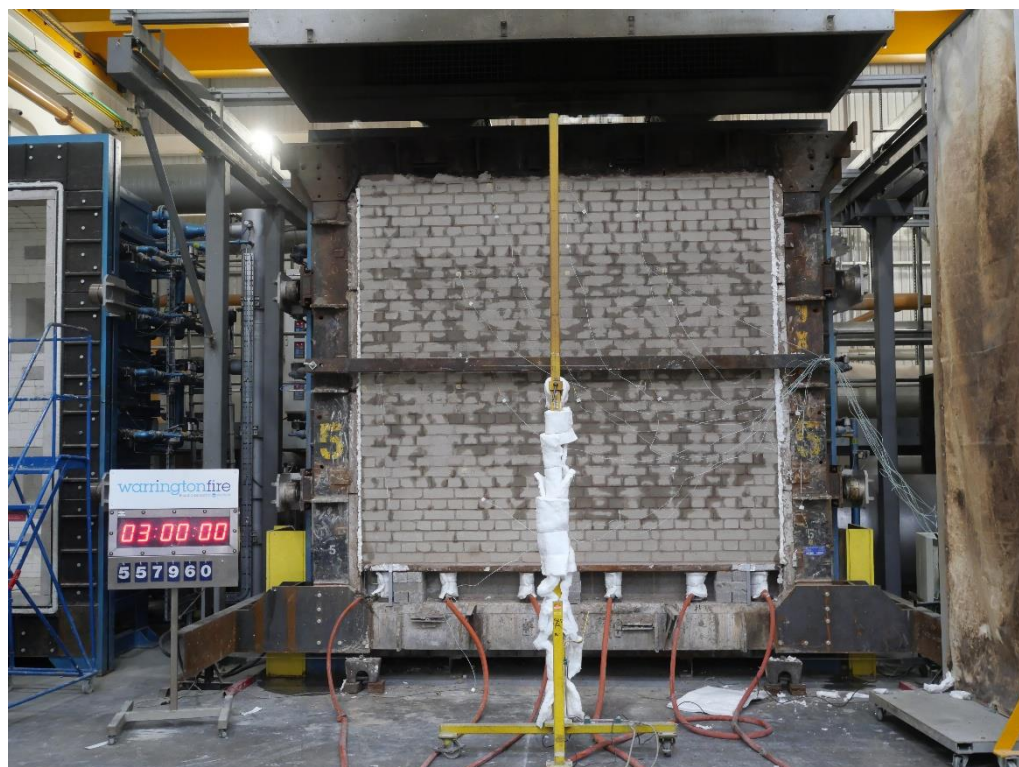
The unexposed face of the wall assembly after a test duration of 60 minutes



The unexposed face of the wall assembly after a test duration of 120 minutes



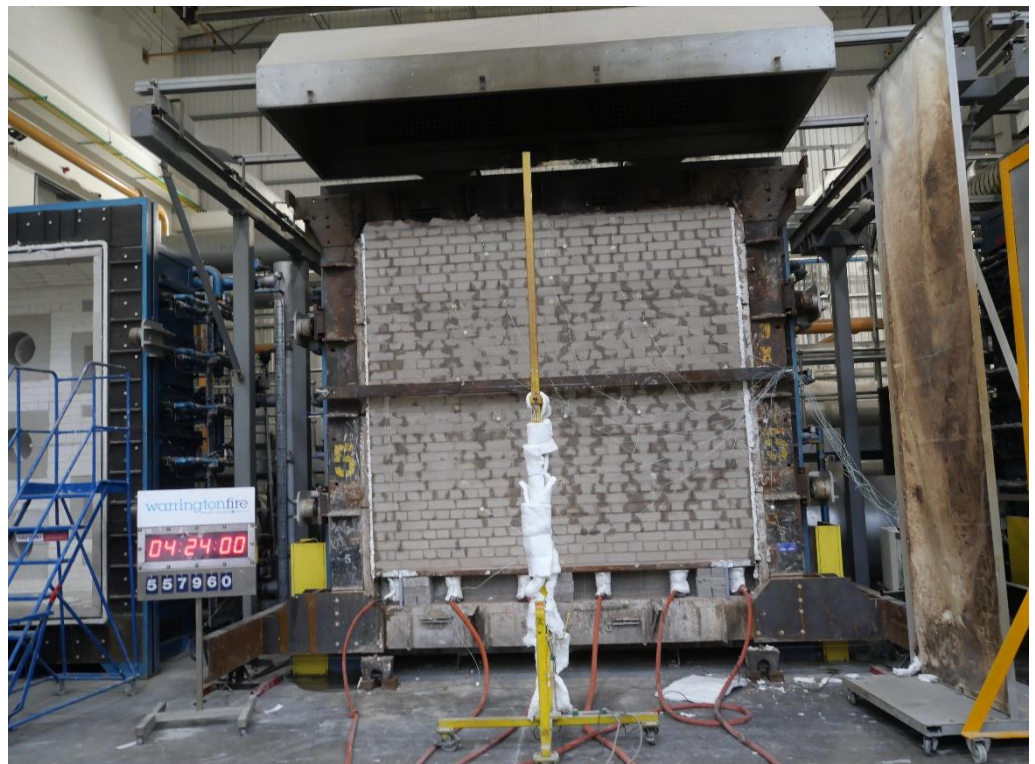
The unexposed face of the wall assembly after a test duration of 180 minutes



The unexposed face of the wall assembly after a test duration of 240 minutes



The unexposed face of the wall assembly at the termination of the test



The exposed face
of the wall
assembly
immediately after
the test



Test Data

Mean furnace temperature, together with the temperature/time relationship specified in BS EN 1363-1: 2020

Time Mins	Specified Furnace Temperature Deg. C	Actual Furnace Temperature Deg. C
0	20	20
8	645	644
16	748	753
24	809	815
32	851	857
40	885	892
48	912	908
56	935	933
64	955	956
72	973	975
80	988	996
88	1003	995
96	1016	1014
104	1028	1030
112	1039	1044
120	1049	1056
128	1059	1058
136	1068	1064
144	1076	1072
152	1084	1080
160	1092	1087
168	1099	1095
176	1106	1102
184	1113	1109
192	1119	1115
200	1126	1122
208	1131	1128
216	1137	1134
224	1142	1139
232	1148	1144
240	1153	1147
248	1158	1153
256	1162	1158
264	1167	1163

Individual and mean temperatures recorded on the unexposed surface of the specimen

Time Mins	T/C Number 20 Deg. C	T/C Number 21 Deg. C	T/C Number 22 Deg. C	T/C Number 23 Deg. C	T/C Number 24 Deg. C	Mean Temp Deg. C
0	17	17	17	17	17	17
8	17	17	17	17	17	17
16	17	17	17	17	17	17
24	17	17	17	17	17	17
32	17	18	17	17	17	17
40	19	22	18	17	18	19
48	25	29	19	18	20	22
56	43	37	23	21	23	29
64	52	45	27	25	29	36
72	57	51	32	32	*	43
80	61	55	36	38	*	48
88	62	58	41	44	*	51
96	62	61	45	49	*	54
104	63	63	49	52	*	57
112	63	64	52	55	*	59
120	63	66	53	57	*	60
128	64	67	56	58	*	61
136	64	68	57	60	*	62
144	65	68	59	61	*	63
152	65	69	61	62	*	64
160	66	69	61	63	*	65
168	67	69	63	64	*	66
176	67	69	64	65	*	66
184	68	70	65	66	*	67
192	68	70	65	67	*	68
200	69	70	66	68	*	68
208	69	71	67	69	*	69
216	70	71	68	70	*	70
224	70	72	69	71	*	71
232	71	73	70	72	*	72
240	71	73	71	73	*	72
248	72	74	72	75	*	73
256	72	75	73	76	*	74
264	73	75	74	77	*	75

* Indicates Thermocouple Malfunction

Individual temperatures recorded on the unexposed surface of the specimen

Time Mins	T/C Number 25 Deg. C	T/C Number 26 Deg. C	T/C Number 27 Deg. C	T/C Number 28 Deg. C
0	17	17	17	17
8	17	17	17	17
16	17	17	17	17
24	17	17	17	17
32	17	17	18	17
40	17	17	18	17
48	18	18	20	18
56	20	20	24	21
64	*	22	30	27
72	*	25	36	36
80	*	29	45	*
88	*	33	56	*
96	*	37	61	*
104	*	40	64	*
112	*	43	67	*
120	*	45	69	*
128	*	47	71	*
136	*	49	72	*
144	*	51	72	*
152	*	53	73	*
160	*	54	74	*
168	*	55	75	*
176	*	57	75	*
184	*	58	75	*
192	*	59	76	*
200	*	60	76	*
208	*	61	77	*
216	*	63	77	*
224	*	64	77	*
232	*	65	78	*
240	*	67	79	*
248	*	69	79	*
256	*	70	80	*
264	*	72	80	*

* Indicates Thermocouple Malfunction

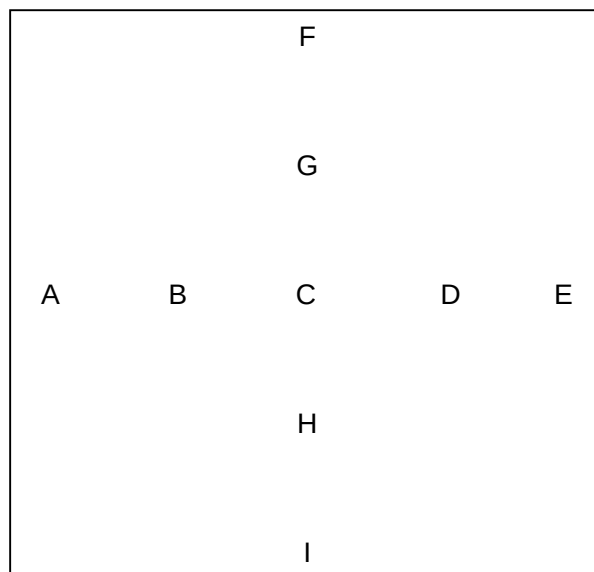
Individual temperatures recorded on the unexposed surface of the specimen

Time	T/C	T/C	T/C	T/C
Mins	Number	Number	Number	Number
	29	30	31	32
	Deg. C	Deg. C	Deg. C	Deg. C
0	17	16	16	17
8	17	17	17	17
16	17	17	21	18
24	17	19	27	20
32	17	21	33	22
40	17	25	39	24
48	18	29	45	26
56	20	33	51	29
64	24	37	56	32
72	31	41	62	37
80	37	46	66	42
88	43	50	70	47
96	48	53	74	52
104	53	56	77	58
112	57	60	79	63
120	61	63	81	68
128	64	65	83	71
136	66	67	85	73
144	68	69	86	75
152	70	70	88	77
160	71	71	89	78
168	72	73	91	79
176	73	73	92	80
184	74	73	93	81
192	74	74	95	82
200	75	75	96	82
208	76	75	97	83
216	77	76	98	84
224	77	76	99	85
232	78	77	100	87
240	78	78	102	88
248	79	78	103	89
256	79	79	105	90
264	79	79	106	92

Recorded heat radiation intensity from the specimen

Time Mins	Radiation Intensity kW/m ²
0	0.000
8	0.013
16	0.009
24	0.013
32	0.001
40	0.014
48	0.021
56	0.033
64	0.043
72	0.058
80	0.077
88	0.085
96	0.091
104	0.110
112	0.103
120	0.121
128	0.115
136	0.128
144	0.131
152	0.132
160	0.148
168	0.138
176	0.147
184	0.143
192	0.169
200	0.164
208	0.170
216	0.168
224	0.177
232	0.180
240	0.197
248	0.186
256	0.187
264	0.201

Deflection of the specimen during the test



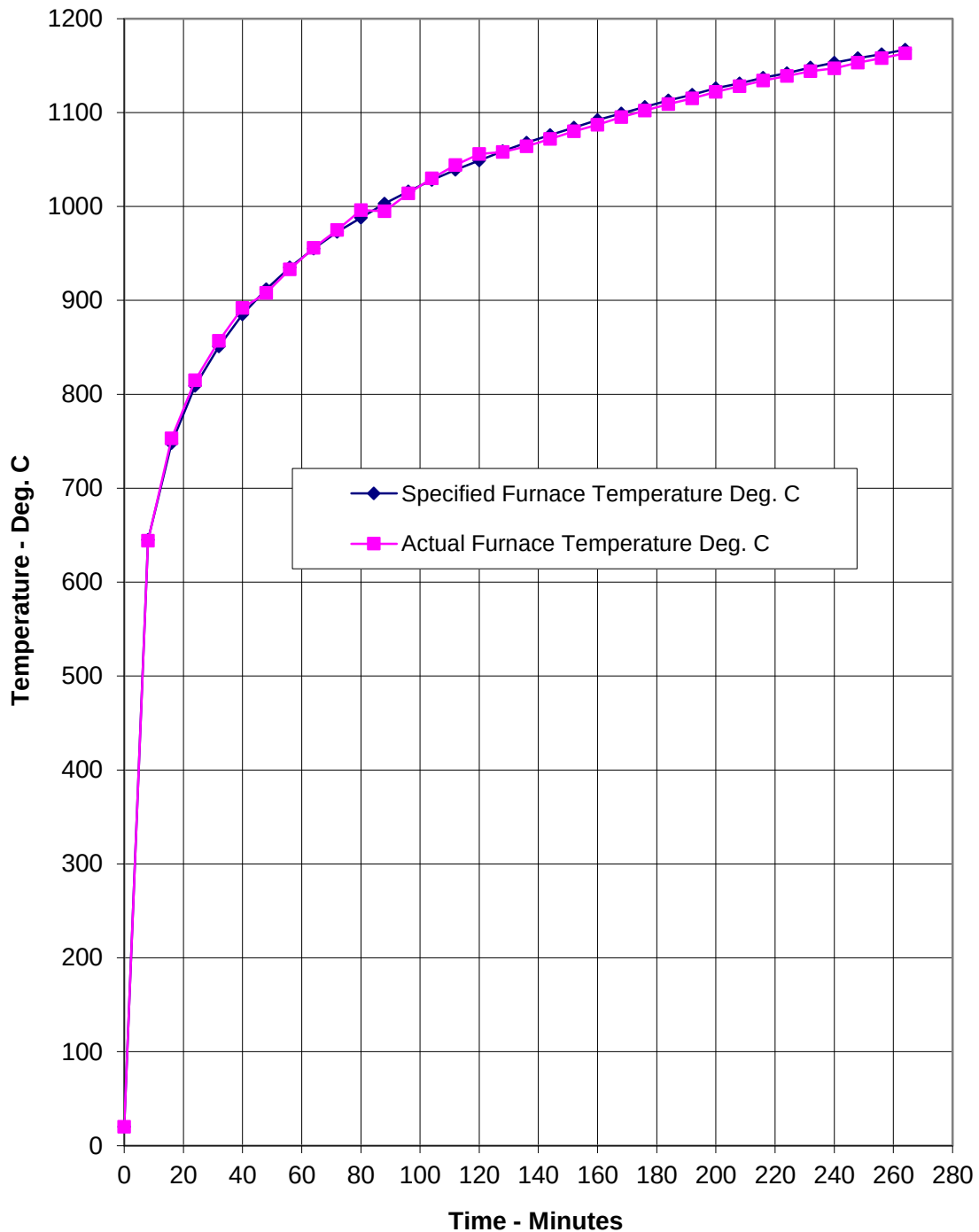
TIME mins	A	B	C	D	E	F	G	H	I
0	0	0	0	0	0	0	0	0	0
10	0	4	7	6	1	1	7	5	1
20	1	8	11	10	3	-1	12	8	1
30	3	9	15	10	9	-2	12	9	2
40	2	9	14	10	-1	1	12	10	2
50	-3	8	15	9	1	-3	9	8	2
60	-2	11	12	9	0	-4	12	9	2
70	-2	9	11	9	-6	-4	11	8	1
80	-3	8	12	9	-3	-2	8	9	2
90	-4	9	11	11	0	-1	10	8	1
100	-1	8	11	8	0	-4	11	6	2
110	-3	6	14	8	-1	-3	10	7	2
120	-1	8	17	9	-1	-2	9	8	2
140	-1	10	14	8	-3	-2	13	9	2
160	-1	9	15	12	0	0	11	9	2
180	0	11	16	13	-1	-5	15	9	2
200	-2	11	17	12	9	-5	12	10	2
220	-3	13	17	13	-3	-3	11	10	2
240	-5	12	17	12	0	3	10	11	3
260	-1	13	19	13	0	-1	14	11	3

Positive deflections indicate movement towards the furnace chamber

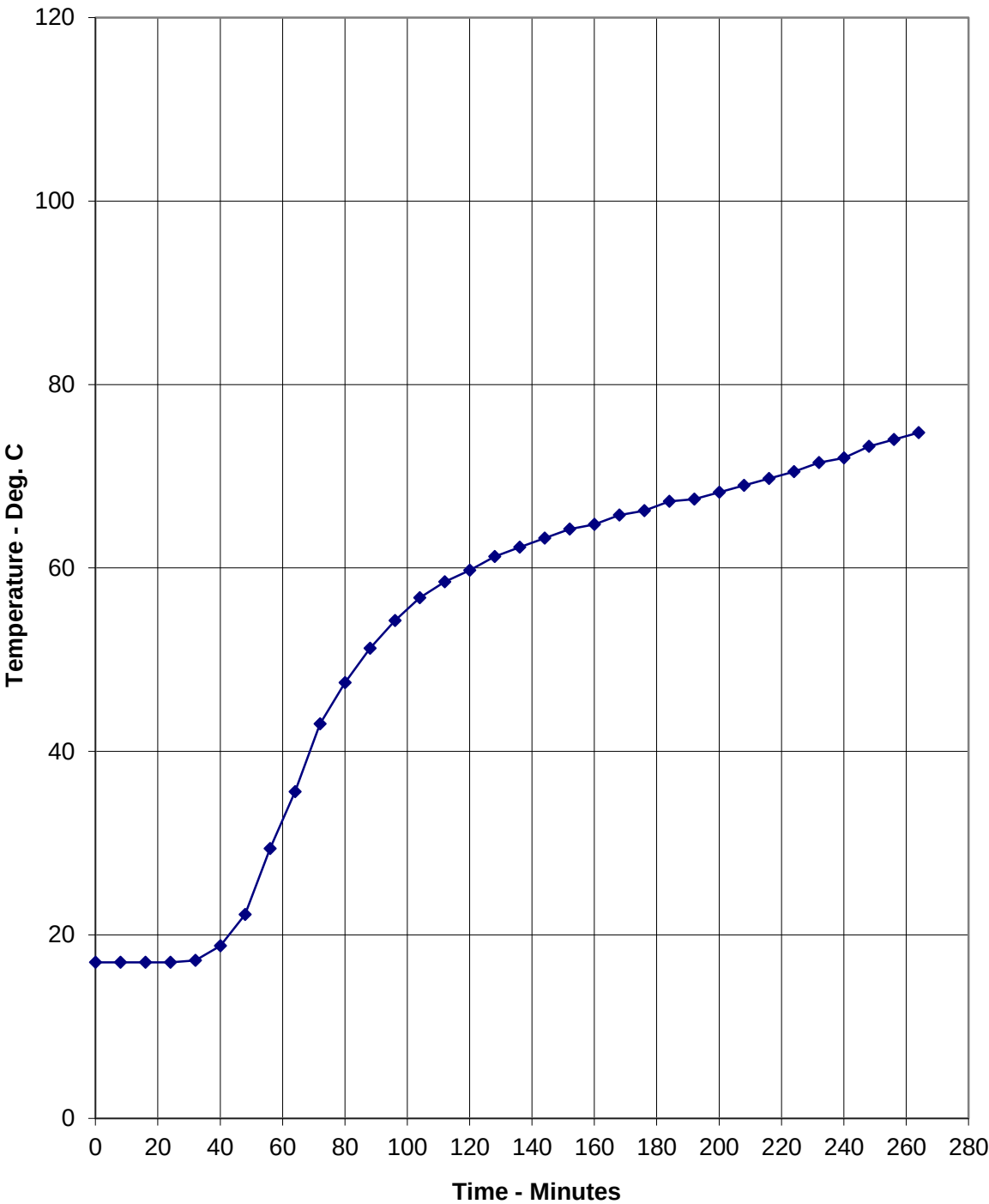
Individual and mean recorded vertical contraction of specimen during the test

Time Mins	Left Vertical Contraction mm	Right Vertical Contraction mm	Mean Vertical Contraction mm	Rate Of Contraction mm/min
0	0.000	0.000	0.000	0.000
8	0.016	0.002	0.009	0.001
16	0.010	0.002	0.006	-0.001
24	-0.003	0.002	-0.001	-0.001
32	-0.007	-0.004	-0.006	-0.001
40	-0.011	-0.024	-0.018	-0.003
48	-0.012	-0.056	-0.034	-0.002
56	-0.014	-0.133	-0.074	-0.002
64	-0.016	-0.168	-0.092	-0.002
72	-0.019	-0.388	-0.204	0.000
80	-0.020	-0.444	-0.232	-0.003
88	-0.070	-0.463	-0.267	-0.001
96	-0.071	-0.470	-0.271	-0.001
104	-0.072	-0.475	-0.274	0.000
112	-0.073	-0.478	-0.276	0.000
120	-0.074	-0.480	-0.277	-0.001
128	-0.075	-0.482	-0.279	0.001
136	-0.075	-0.483	-0.279	0.000
144	-0.075	-0.483	-0.279	0.000
152	-0.075	-0.483	-0.279	0.000
160	-0.075	-0.484	-0.280	0.000
168	-0.075	-0.484	-0.280	0.000
176	-0.075	-0.485	-0.280	0.000
184	-0.076	-0.484	-0.280	0.001
192	-0.076	-0.485	-0.281	0.000
200	-0.076	-0.485	-0.281	0.000
208	-0.076	-0.485	-0.281	0.000
216	-0.076	-0.486	-0.281	-0.001
224	-0.076	-0.486	-0.281	0.000
232	-0.076	-0.487	-0.282	-0.001
240	-0.077	-0.487	-0.282	-0.001
248	-0.076	-0.487	-0.282	0.000
256	-0.077	-0.488	-0.283	0.000
264	-0.077	-0.488	-0.283	0.000

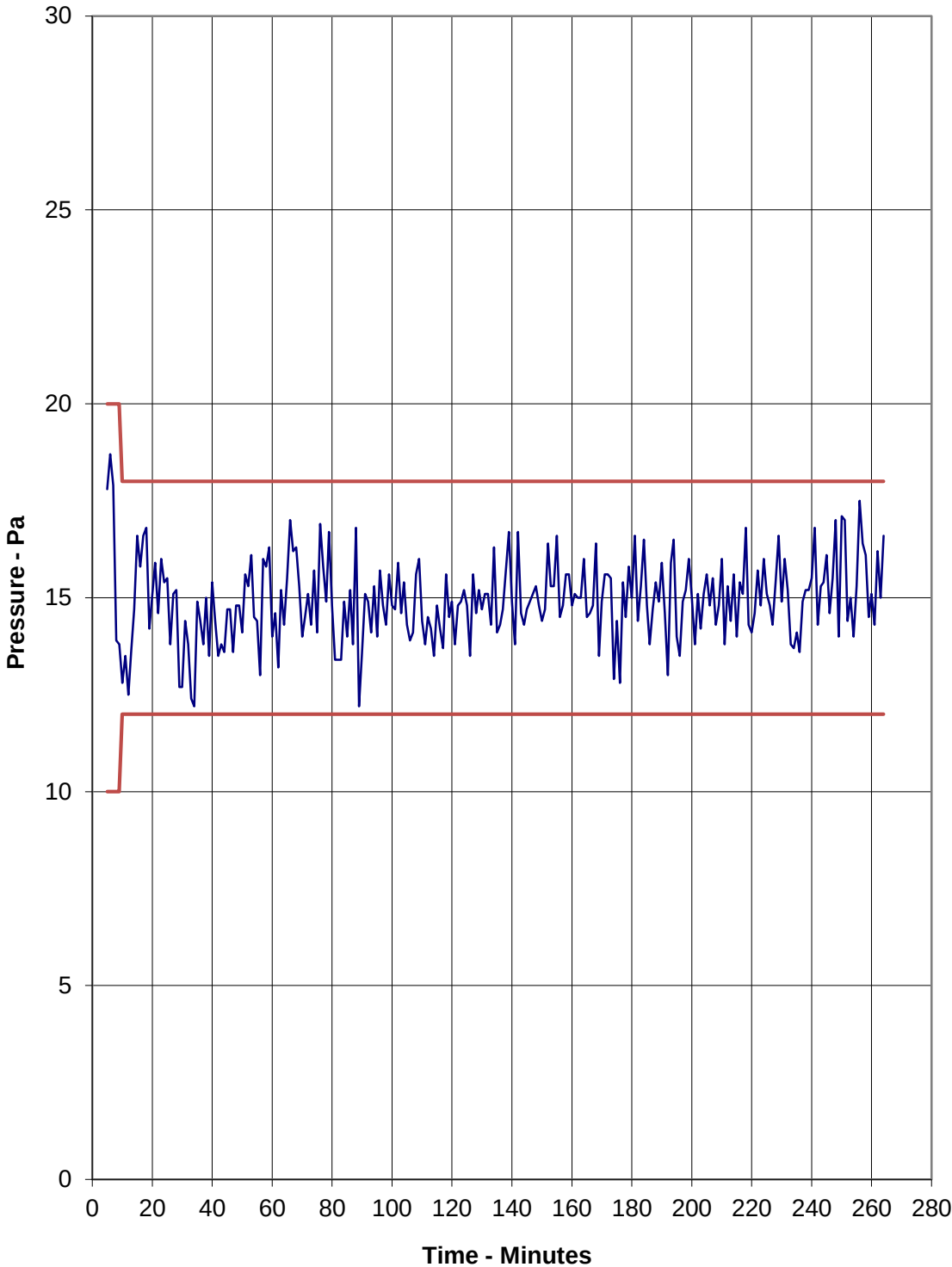
Graph showing mean furnace temperature, together with the temperature/time relationship specified in BS EN 1363-1: 2020



Graph showing mean temperature recorded on the unexposed surface of the specimen



Graph showing recorded furnace pressure 500 mm below the head of the wall assembly



On-going Implications

Validity

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The test results relate to the behaviour of the test specimens of a product under the particular conditions of the test; they are not intended to be the sole criteria for assessing the potential fire hazard of the product in use, nor can the results be extrapolated and applied to other products.

Reports are statements of fact(s) prepared in accordance with the referenced version of the standard(s). Reports are based upon the information provided to Warringtonfire. Warringtonfire takes no responsibility for the accuracy or completeness of such information.

The results stated in this report apply to the test specimens as received. The results relate only to the behaviour of the specimens of the element of construction under the particular conditions of test. They are not intended to be the sole criteria for assessing the potential fire performance of the element in use, nor do they reflect the actual behaviour in fires.

This report details the method of construction, the test conditions and the results obtained when the specific element of construction described herein was tested following the procedure outlined in BS EN 1365-1:2012, BS EN 1363-1:2020, and where appropriate BS EN 1363-2:1999.

Any significant deviation with respect to size, constructional details, loads, stresses, edge or end conditions other than those allowed under the field of direct application in the relevant test method is not covered by this report.

Any differences in relation to the aforementioned characteristics may significantly affect the performance and will therefore invalidate the application of the test results to the variant product. It is recommended that any proposed variation to the tested configuration or product should be referred to the test sponsor. The test sponsor should then obtain appropriate documentary evidence of compliance from Warringtonfire or another accredited testing authority. The supplier of the product is responsible for ensuring that the product which is supplied for use is identical to the test specimens that were tested.

The specification and the interpretation of fire test methods are both the subject of ongoing development and refinement. Changes in the applicability of the results of tests in relation to associated legislation may also occur. For these reasons the currency and the relevance of test reports should be considered by the user.

The test report also relates only to the sample(s) of the product submitted to the test. The laboratory accepts no responsibility for the representativeness of the test specimens unless so stated in the test report.

Confidence that the product that is supplied to the market will have the performance indicated in the test report can be supported by use of third-party certification schemes.

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Uncertainty of measurements

Because of the nature of fire resistance testing and the consequent difficulty in quantifying the uncertainty of measurement of fire resistance, it is not possible to provide a stated degree of accuracy of the result.

EGOLF

Certain aspects of some fire test specifications are open to different interpretations. EGOLF have identified a number of such areas and have agreed Resolutions which define common agreement of interpretations between fire test laboratories which are members of the Groups. Where such Resolutions are applicable to this test they have been followed:

020-2018 - Average temperature thermocouples moved away from stiffeners, joints, through components, features and hotspots

025-2018 - Procedures for temperature measurement in fire resistance tests

028-2018 - Disregarding unexposed thermocouples

050-2018 - Deduction of the insulation performance criteria

061-2021 - Harmonization of pressure measurement, processing and reporting

062-2022 V2 - Reporting test results

Revision history N/A

Note: The field of direct application may only be defined following the identification of classification(s). The field of direct and, where applicable, extended application will be included in the classification report.

Sample Report

Not applicable



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